

Revision Rotator Cuff Repair

Joseph D. Cooper, Anthony Essilfie, Seth C. Gamradt

Primary rotator cuff repair frequently leads to successful subjective results with decreased pain and increased function, but incomplete healing or retearing is known to commonly occur.¹⁻⁵ Retear rates following repair of small to medium tears have been reported as high as 40%, and up to 94% following massive, chronic cuff tears.^{6,7} Despite retear, many patients can maintain pain relief and improved functional outcome.⁸ However, patients who do not heal, on average, tend to have worse satisfaction, clinical outcomes, and decreased function compared to those with an intact repair.^{9,10} Of those patients with retear who are unsatisfied and have continued symptoms, 6% to 8% will require revision rotator cuff repair.^{11,12} This chapter provides an approach to patient evaluation after unsatisfactory rotator cuff repair and details surgical options for this difficult problem.

HISTORY

A careful history is critical in understanding the etiology of rotator cuff repair failure and to understand the patient's expectations. Typically, patients with failed rotator cuff repairs present with shoulder pain and weakness. It is important to ascertain the onset, location, characterization, intensity of the pain, and the patient's physical demands for both work and recreation. Furthermore, it is important to elucidate whether any specific motions or activities elicit the pain. The surgeon should ask if there was a period of improvement after the primary surgery or if the symptoms continued to persist since the index procedure. The indication for most revision rotator cuff repairs is usually to manage persistent pain and weakness since the initial surgery.^{13,14} All previous surgeries should be discussed with the patient; obtaining previous medical records, operative reports, preoperative imaging, and arthroscopic pictures are helpful in assisting with treatment decisions.

When considering a revision procedure, the most important factor to ascertain from the history is the possible cause of failure—the exact cause of which is often multifactorial. Factors causing rotator cuff nonhealing or retear can be classified into three broad categories: biologic, technical, and traumatic. Injured tendons develop a fibrovascular scar at the tendon bone interface, which is altered from the native tissue with decreased vascularity and decreased healing potential.¹⁵ This biologically inferior milieu predisposes a repaired tendon to retearing.

Biologic

This category of risk factors includes both the biology and the nature of the cuff tear itself, as well as the patient as a whole. Patient factors, such as increased age, nicotine use, and diabetes, have all been consistently reported to lead to decreased healing rates and poor clinical outcomes.^{7,15-19} Preoperative factors related to the cuff tear itself, including tear size, degree of retraction, and muscle atrophy, have all been shown to correlate with increased failure rates.^{16,20} As the cuff ages, and the longer the tear has persisted, there is an increased level of fatty infiltration with decreased muscle in the cuff. Studies have shown that fatty infiltration and cuff atrophy can lead to increased tension on the repair and increased failure rates.^{21,22}

Infection is an uncommon cause of cuff repair failure; however, it must be ruled out. Patients who have persistent postoperative pain with no period of relief, who have compromised immune systems, a history of fevers, redness, or problems with their incision should undergo an infection work-up consistent with white blood cell count, erythrocyte sedimentation rate blood test, c-reactive protein test, joint aspiration for cell count, and culture.

Technical Factors

Identifying potential surgical errors as a cause of failure can be essential when planning a revision. These factors can include knot failure, inadequate recognition of tear size or pattern, poorly fixed or positioned anchors, and over-tensioning the repair. One of the most common modes of failure for rotator cuff repair is when sutures cutout through poor quality tendon^{23,24} Suture configurations, such as double row, single row, transosseous, and transosseous equivalent repairs, have been described to address this problem. Biomechanical data support double row fixation and/or transosseous equivalent repair,²⁵⁻²⁹ but clinical data have not consistently shown improved outcomes over a single row repair.³⁰⁻³² Surgeon familiarity with the chosen technique is likely most essential, as low surgeon volume has been shown to be an independent risk factor for reoperation.¹⁸

Failure to address other pain generators in the shoulder can also lead to persistent pain and decreased function, which can be incorrectly attributed to a failed repair. Failure to address biceps tendon pathology, prominent acromial spurs, or a symptomatic

Abstract

Revision rotator cuff repair is a challenging orthopaedic problem that is becoming increasingly more frequent. This chapter will review the treatment of revision rotator cuff tears including history, physical exam, imaging, decision-making, and treatment.

Keywords

rotator cuff revision
shoulder
sports
revision
rotator cuff

TABLE 49.1 Noncuff Tear Etiologies of Residual Shoulder Pain**Extrinsic Pain Generators**

- Cervical radiculopathy
- Referred pain from intrathoracic or intra-abdominal pathology
- Suprascapular neuropathy

Intrinsic Pain Generators

- Os acromiale
- Instability
- Biceps tendinopathy
- Acromioclavicular joint pain
- Labral tear
- Adhesive capsulitis
- Subacromial impingement
- Glenohumeral arthritis

acromioclavicular joint at the index surgery can contribute to a persistently symptomatic shoulder after rotator cuff repair. Other noncuff tear etiologies of shoulder pain are shown in Table 49.1.

Traumatic

Traumatic tearing can be defined as early (before tendon healing has occur) or late (after healing has occurred). Early tears usually occur between 6 and 26 weeks postoperatively³³ and can be a result of overly aggressive physical therapy in the acute recovery period. Late traumatic retears usually have a history of improved or return to normal shoulder function, followed by an identifiable trauma or injury that led to onset of new symptoms.

PHYSICAL EXAMINATION

The physical exam is paramount in identifying the cause of a failed rotator cuff repair and also the suitability of the patient for a second surgery. We start our evaluation with a cervical spine examination. Many patients with cervical spine disease have associated shoulder pain. They usually have limited cervical range of motion (ROM) particularly with hyperextension. It is important to note if there is any decreased sensation in a particular dermatome. A Spurling test should be performed to evaluate for radicular symptoms.

Examination of the shoulder begins with inspection of the shoulder by looking for evidence of supraspinatus or infraspinatus atrophy. Also assessing previous portals or incisions should be noted for healing, contracture, or signs of infection. The surgeon should palpate the glenohumeral joint, rotator cuff insertion, biceps groove, and acromioclavicular joint. Active and passive ROM of the shoulder should be evaluated and compared to the contralateral shoulder. Often passive ROM is greater than active ROM in the setting of a rotator cuff tear. Patients that have equal loss of passive and active ROM (true stiffness) without deficits in strength may have postoperative adhesions, glenohumeral arthritis, or capsular contracture. The strength of each rotator cuff tendon should be evaluated systematically. Having the patient internally rotate the arm and elevate it 90 degrees

in the scapular plane against resistance tests the supraspinatus. The infraspinatus is tested with resisted external rotation with the arm at the side and elbow with 90 degrees of flexion. The lift-off test and belly press test are done to assess the strength of the subscapularis. The subscapularis also can be assessed with a bear hug test where the patient places the ipsilateral hand on the contralateral shoulder and the elbow positioned anterior to the body. The patient then actively resists the examiner's attempt externally rotate the forearm. If the patient is unable to keep his hand on his shoulder or weakness is detected, the test is positive.³⁴

Neer and Hawkins signs are used to evaluate for subacromial impingement. A hornblower sign is representative of significant deficit in the posterior rotator cuff (infraspinatus and teres minor). These patients commonly abduct the arm when trying to bring their hand to their face with their arm at the side. The cross-body adduction test is used to evaluate for AC joint pathology. Yergason and Speed test assess for biceps pathology.

When it is difficult to discern between cervical radiculopathy versus true shoulder pain, a diagnostic injection in the subacromial space can be helpful.

IMAGING

Standard radiographs of the shoulder including the anteroposterior, Grashey, scapular Y, and axillary views of the shoulder should be taken. When evaluating the radiographs, it is critical to evaluate the glenohumeral joint for degenerative changes or evidence of superior migration of the humeral head as these radiographic changes are most often contraindications to revision rotator cuff repair.

Magnetic resonance imaging (MRI) is probably most important in the work-up of symptomatic patients after a previous rotator cuff repair. It has been shown that MRI is accurate in diagnosing recurrent full thickness rotator cuff tears but is inaccurate in defining tear size.³⁵ MRI has a reported 85.5% sensitivity and 90.4% specificity for all tears.³⁶ There is conflicting evidence assessing whether there is any benefit of a magnetic resonance arthrogram (MRA) versus a noncontrast MRI.³⁷⁻³⁹ Additionally, there is a report of high false-positive results with MRA when used for evaluation for retears.⁴⁰ In the event that a patient has a contraindication for an MRI, computed tomography arthrography is a viable substitute.

Ultrasonography, when performed by an experienced technician, is comparable to MRI in diagnosing and assessing tear size and retraction of primary rotator cuff tears.⁴¹ Ultrasonography has a high sensitivity of 91%, specificity of 86%, and accuracy of 89% for detecting rotator cuff tears postoperatively.⁴² However, MRI is often more useful to surgeons than ultrasound since it allows the ability to assess the shoulder in three planes. When assessing the reparability of a tear, it is important to look at the degree of retraction on the coronal view and the degree of muscle atrophy on the sagittal view.

DECISION-MAKING PRINCIPLES

Factors to consider when determining treatment options of these patients include tear size, degree of retraction, quality of the

rotator cuff muscle, the presence or absence of glenohumeral arthritis, and patient function. Rotator cuff repairs are most likely to fail when the tendon has Goutallier grade 3 or grade 4 fatty degeneration⁴³; muscle atrophy that is greater than grade 3 is a relative contraindication to revision.⁴⁴

An extensive trial of nonoperative management should be exhausted before considering revision repair, as even in the setting of a failed repair radiographically, acceptable clinical and subjective patient outcomes can be achieved with therapy.¹²

Management of asymptomatic patients with a known retear on imaging is challenging. Jost et al.⁴⁵ followed 20 patients that were found to have a rerupture on MRI. These patients were evaluated on average at 3.2 years and again at 7.6 years after the index procedure. Despite retears seen on MRI, there was no significant decrease in Constant score, and the size of the tears did not significantly increase between the two follow-up time points. However, there was significant progression of fatty infiltration of the infraspinatus and the acromiohumeral distance, making future surgeries, if required, significantly less likely to be successful when treated in this delayed fashion. Further research in this cohort is needed to identify factors that lead to symptomatic tears.

Previous literature has shown that ideal candidates for revision rotator cuff repair have an intact deltoid origin, with preoperative active elevation of 90 degrees and forward flexion of 120 degrees.⁴ There should be good tissue quality, adequate bone stock, and the apex of the tendon should not have retracted beyond the glenohumeral joint. There should be no signs of active infection, glenohumeral arthritis, or cuff tear arthropathy (Fig. 49.1).

Prior to considering revision surgery there should be a candid discussion and understanding between patient and surgeon regarding the reason for revision and the expected outcomes (Table 49.2).

TREATMENT OPTIONS

Nonoperative

As discussed in the previous section, a trial of physical therapy and nonoperative management should be exhausted before revision surgery in most patients. The only exception being physiologically young active patients with an acute traumatic retear. These cases should be fixed acutely. Therapy focusing on periscapular strengthening and the anterior deltoid has been shown to be most effective at helping restore the force couple of a failed rotator cuff.⁴⁶

Débridement Alone

In elderly, low-demand patients with persistent pain, but preserved ROM, arthroscopic débridement with a biceps tenotomy or tenodesis can be considered.¹ The biceps tendon is a pain generator and can be released without loss of shoulder function. Short-term outcomes have shown that débridement alone can successfully relieve pain in shoulders with massive cuff tears but no significant arthritis.^{47,48}

TABLE 49.2 Decision-Making Prior to Revision Cuff Repair

Is This Patient a Candidate?

- Patient factors:
 - Age
 - Function/demand level
 - Overall health status
 - Number of prior surgeries
 - Pain level
- Cuff/shoulder factors:
 - Preoperative range of motion
 - Status of the deltoid
 - Level of fatty degeneration
 - Tendon retraction
 - Other pain generators causing problems
 - Arthritis

Has There Been Sufficient Nonoperative Management?

- Duration of therapy
- Adherence to prescribed therapy program
- Effectiveness of prescribed therapy program (anterior deltoid focused)
- Nonsteroidal antiinflammatory drugs

Does the Patient Have Appropriate Expectations/Understanding?

- Patient should understand:
 - Revision surgery is primarily for pain relief
 - Functional improvements cannot be promised
 - Retearing (again) is common
 - Will not result in a “normal” shoulder

Revision Repair

Once it is determined that the patient is a candidate for revision cuff repair, planning the revision procedure begins with evaluation of the failed procedure. Open versus arthroscopic repair options are based on surgeon familiarity and preference, with current repair trends largely biasing arthroscopic repair.

In modern rotator cuff repair, biologically absorbable screws are frequently used. If metal anchors were used these should only be removed if prominent, because removal causes bony defects that weaken the greater tuberosity or fixation.¹ New anchors should be tapped to prevent damaging them on retained hardware. When possible, oversized hardware should be used at prior hardware sites to avoid further bone disruption. An alternative is to perform a transosseous repair and avoid new anchor placement.¹

In revision rotator cuff repair the tendons are often retracted and scarred, requiring various mobilization techniques and soft tissue releases to free the tissues from adhesions and allow a tension-free anatomic repair. Multiple interval slide techniques have been described for open surgery⁴⁹⁻⁵⁴ and have been adapted for arthroscopic use. An anterior interval slide, oriented towards the base of the coracoid releasing the interval between the supraspinatus and the rotator interval, can achieve 1 to 2 cm of lateral excursion.⁵⁵ A posterior interval slide releases between the supraspinatus and the infraspinatus and provides 1 to 2 cm of lateral excursion; however, the suprascapular nerve is at risk with this

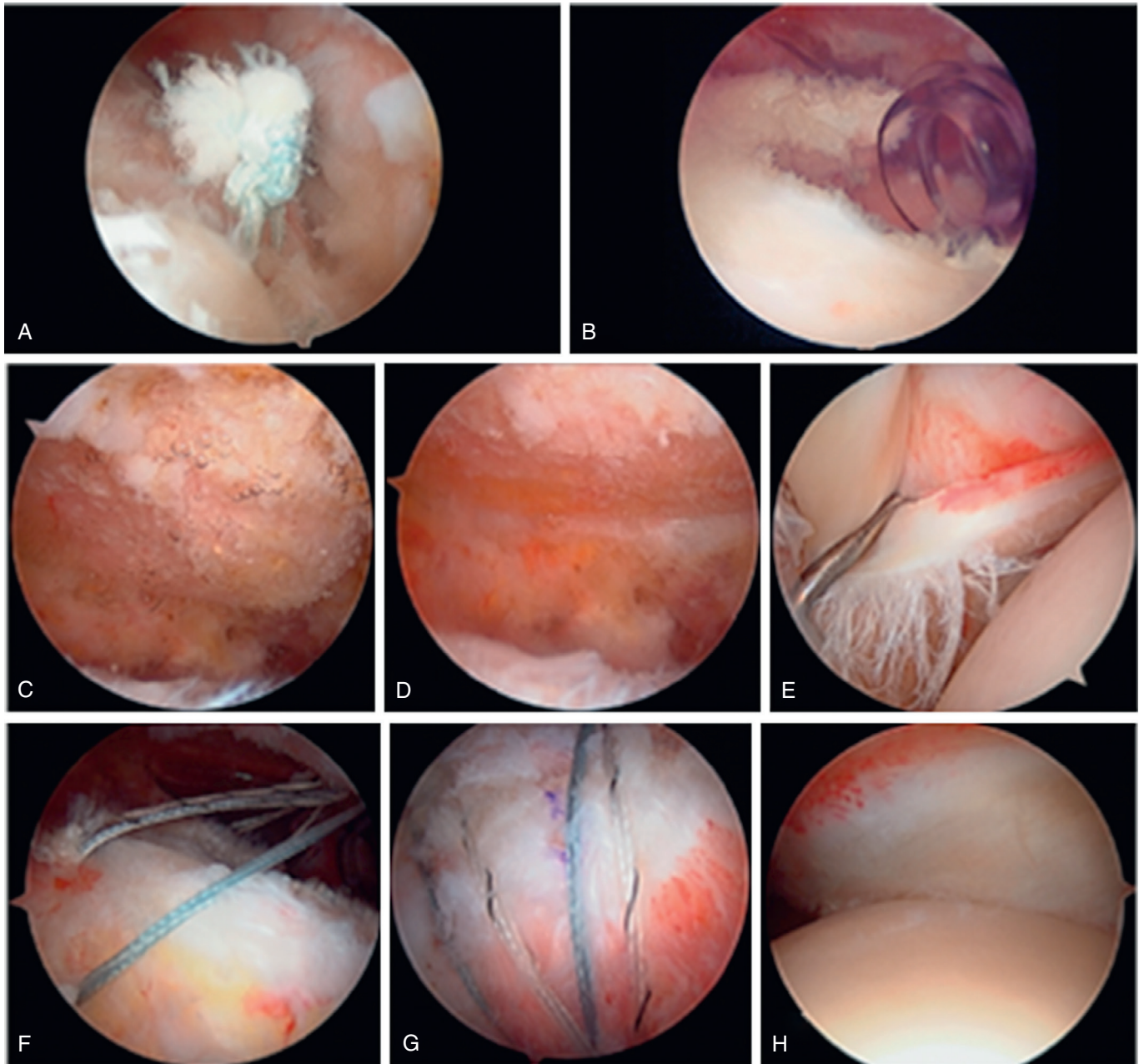


Fig. 49.1 Arthroscopic images demonstrate concomitant pathology in revision rotator cuff repair in a 55-year-old man who is a surfer. (A) View shows simple sutures pulled through the lateral cuff. (B) Views show a 2x2 cm supraspinatus tear following débridement of suture material and nonviable rotator cuff. (C) View shows residual acromial spur. (D) View obtained following acromioplasty. (E) View shows biceps tear and severe tendinitis that necessitated biceps tenodesis. (F) View of medial row sutures. (G) Subacromial view of a double-row revision repair. (H) Intra-articular view of completed rotator cuff repair. (Reproduced with permission from Johnson BC, Dunphy TR, Gamradt SC. Revision Rotator Cuff Repair. In: Gamradt SC, Sperling, JW, Galatz LM, eds. *Let's Discuss Revision Shoulder*. Rosemont, IL: AAOS; 2015:23–41.)

release. These procedures can be combined in a double interval slide and achieve up to 5 cm of additional laterality for a tension-free repair.⁵⁵

When the tendons are mobilized sufficiently to reach the tuberosity they can be fixed to the humerus with anchors. As previously mentioned, double row and transosseous-equivalent repairs have been shown to be biomechanically superior to a single row in multiple studies^{25–29}; however, no study has yet

shown a clinically significant difference in outcomes following primary rotator cuff repair (RTC) repairs.^{30–32} This has not been specifically studied in revision procedures. However, in the revision setting with compromised tissue the theoretical benefit of a stronger repair is likely warranted, and a double row repair should be used when possible.

Burkhart⁵⁶ described a partial RTC repair technique for irreparable tears when the tendon cannot be fully mobilized to the

tuberosities. This technique restores the force couple of the RTC by repairing the inferior half of the infraspinatus and the whole subscapularis. Multiple modifications of this technique have since been described.⁵⁷ These techniques have up to a 41.7% retear rate and in the modern treatment of failed RTCs have largely given way to reverse total shoulder arthroplasty or other biologic repair techniques such as patch augmentation.

Repair With Biologic Augmentation

Poor patient biology is a frequent cause for primary RTC failure, whether that be muscle atrophy, tendon or bone quality, or tendon retraction. In the revision setting, surgeons frequently look to augment the biologic milieu of the shoulder to aid in healing. The use of growth factors, stem cell therapy, gene therapy, and bioimplants have all been described, although currently mostly in animal models.⁵⁸ The most frequently used technique involves stimulating marrow elements from the humeral head with a microfracture technique with or without the application of a biologic patch based on the integrity of the tissue. Marrow elements provide polymorphonuclear cells, macrophages, and a fibrin clot (acts as a temporary scaffold), as well as growth factors and platelets to aid in cuff healing.⁵⁹ Multiple patch types have been described including degradable or nondegradable, synthetic grafts, xenografts, and extracellular-matrix-based patches.⁶⁰⁻⁶⁴ The patch can be used as an additional support for tendons with poor quality, and sewn over the repair.^{64,65}

Superior capsular reconstruction has been described by Mihata and colleagues for massive irreparable rotator cuff tears.⁶⁶ This technique uses a graft that spans from the superior glenoid to the greater tuberosity. Early outcomes in primary RTC repairs have shown it to reliably decrease pain and to provide some return of shoulder function.⁶⁷ This technique has not been specifically evaluated in regard to revision RTC procedures, but it may be an option for patients with minimal arthritis trying to avoid reverse total shoulder arthroplasty.

Other biologic additives have also been described and are used with anecdotal success; however, their outcomes have not been clinically proven in the literature.⁵⁸

For Tears That Cannot Be Repaired

Massive revision cuff tears that cannot be mobilized enough to restore the RTC tendons to their native position have to be repaired by alternative means. Patients with preserved joint space but limited ROM are candidates for tendon transfers. Both latissimus dorsi and trapezius transfers have been described.^{68,69} These procedures can successfully restore joint motion and are a reasonable option for the young patient who does not want to undergo shoulder arthroplasty. Reverse total shoulder arthroplasty is the treatment of choice for elderly patients with tears that cannot be repaired and who have significant glenohumeral arthritis.

POSTOPERATIVE MANAGEMENT

Patient immobilization is based on tear size and quality. Large tears are fully immobilized for 4 weeks postoperatively. In smaller tears we allow shoulder shrugs, passive external rotation to neutral, and passive forward elevation to 90 degrees. No large studies have evaluated the optimal rehabilitation strategies after revision rotator cuff repair, and management is largely based on surgeon preference and experience. We advise patients that pain and functional improvement can continue for 1 year or longer.

RESULTS AND COMPLICATIONS

Revision rotator cuff surgery can result in reduced pain with improved shoulder function and patient satisfaction.³ Multiple studies have now supported this and are summarized in [Table 49.3](#).

While revision RTC repair does generally lead to pain relief and improved function, it does have a higher complication rate and higher failure rate than primary repair. The retear rate



Authors' Preferred Technique

We offer revision RTC repair to patients with acceptable tissue quality on MRI, no pseudoparalysis, a functioning deltoid, and only one previous repair. Patients are counseled preoperatively that retearing is common, pain relief and restoration of function may not be complete, and the durability and long-term results of revision rotator cuff repair are not well studied.

We use only arthroscopic techniques for revision cuff repair. This avoids the risks associated with splitting the deltoid or other deltoid takedown techniques required for an open approach. The patient is positioned in the beach chair position with the arm in a pneumatic arm holder that facilitates subacromial distraction and allows rotational positioning of the arm. Epinephrine in the arthroscopy fluid at of 1 mL of 1:1000 per 3000 mL of saline improves hemostasis and visualization.

Arthroscopic portals from the initial procedure should not influence portal placement for the revision procedure. We start with a posterior viewing portal placed at 1 cm inferior and 1 cm medial to the posterolateral acromion (slightly more superolateral than is standard). This position is not ideal for glenohumeral access but provides better visualization of the subacromial space. Anterior and lateral portals are placed under direct vision based on access to required working

sites. Tear mobilization for a tension-free repair is frequently required. Articular sided mobilization is performed with radiofrequency ablation by releasing adhesions between the cuff and the labrum. This release is limited to 2 cm medial to the glenoid to avoid suprascapular nerve injury. A bursal-sided release can be carried anteriorly to the coracoid and as far posteriorly as necessary, often to the scapular spine. During revision bursectomy, scarring can make differentiation of the cuff tissue from bursa difficult; the cuff can be protected by starting from normal tissue anteriorly and carrying the bursectomy posteriorly with the shaver aimed cephalad. In anterosuperior cuff tears, an anterior interval slide, or a release of the thickened medial sling between the subscapularis and the supraspinatus can be performed without affecting the failure rates.⁷⁰

Vessels are frequently encountered during the posteromedial release and bleeding can lead to a "redout" situation obscuring the camera. This can be managed by keeping the site of bleeding directly in the camera field of vision using the pressure of the water to slow the bleeding then moving the cautery device to the arthroscope and cauterizing the source.

Repairs are performed using bioabsorbable anchors in a double row formation when possible ([Fig. 49.2](#)).

Continued



Authors' Preferred Technique—cont'd

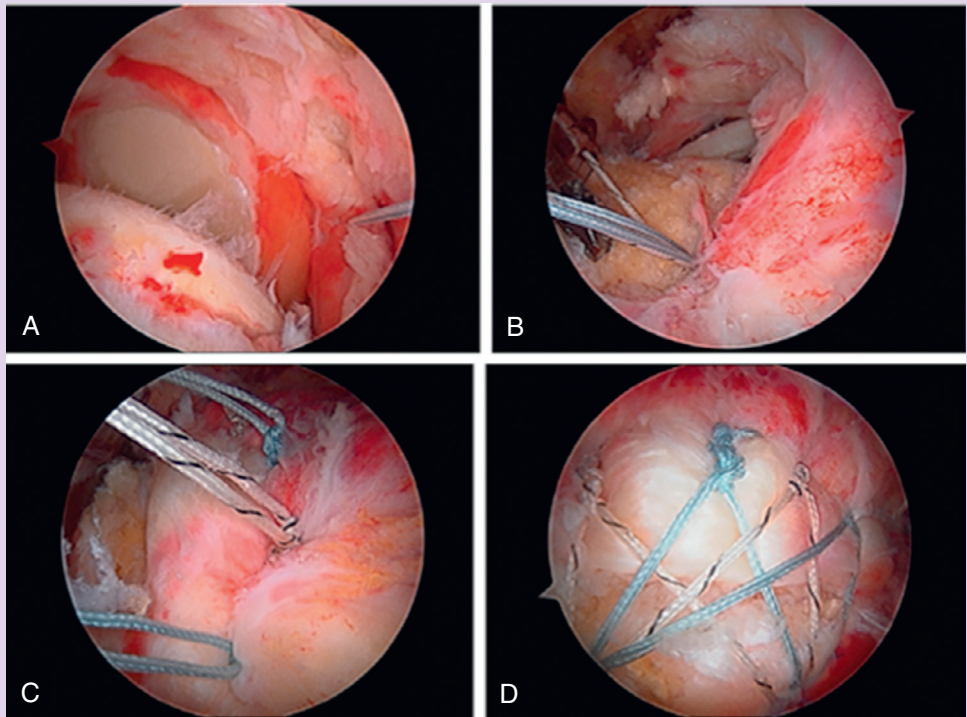


Fig. 49.2 Arthroscopic views of the shoulder in a 62-year-old man who is a dock worker, with weakness 6 months after rotator cuff repair who underwent revision of a massive tear with grade 2 atrophy seen on MRI. (A) View obtained from the lateral portal before mobilization. (B) View obtained from the posterior portal following bursal and articular release. (C) View shows medial-row sutures. (D) View of completed double-row revision repair. *MRI*, Magnetic resonance imaging. (Reproduced with permission from Johnson BC, Dunphy TR, Gamradt SC. Revision Rotator Cuff Repair. In: Gamradt SC, Sperling, JW, Galatz LM, eds. *Let's Discuss Revision Shoulder*. Rosemont, IL: AAOS; 2015:23–41.)

TABLE 49.3 Revision Rotator Cuff Repair Studies

Authors	Year of Publication	No. of Shoulders	Mean Age (Years)	Average Follow-Up in Months	Massive Tears (% of Total)	Design
A. Outcomes						
Mora et al. ⁷²	2017	51	60	25	59%	Retrospective review of prospective data
Shamsudin et al. ⁷¹	2015	50	63	35	NA	Retrospective review of prospective data
Chuang et al. ⁷³	2014	32	69	70	59%	Retrospective case series
Parnes et al. ²	2013	94	58	No average reported, minimum 12	54%	Retrospective case series
Lädermann et al. ⁷⁴	2012	74	61	59	72%	Retrospective case series
Keener et al. ¹³	2010	21	56	36	NA	Retrospective case series
Piasecki et al. ⁷⁵	2010	54	55	31	7%	Retrospective case series

TABLE 49.3 Revision Rotator Cuff Repair Studies—cont'd

Authors	Active Postop Elevation (Gain)	Clinical Outcomes (Gain)	Retear Rate	Complications/Revisions (%)
B. Outcomes				
Mora et al. ⁷²	135.6° (39)	CS: 69.1 (26.4) SST: 7.58 (3.6) VAS: 6.49 (3.2) Unsatisfactory outcome: 27%	60%	15.7% revision rate
Shamsudin et al. ⁷¹	NA (2)	4 Newton increase in abduction strength, no improvement in lift-off strength	40% at 2 years	12% revision rate
Chuang et al. ⁷³	156° (9)	ASES: 87 (NA) UCLA: 30 (14) VAS: 0.9 (3.7)	NA	NA
Parnes et al. ²	NA	NA	NA	Failure to heal: 10.6% Stiffness: 7.4% Infection: 2.1% Nerve injury: 1% 8% Revision rate
Lädermann et al. ⁷⁴	152° (16)	ASES: 77 (26) UCLA: 27 (9) VAS: 2 (3) Pt satisfaction: 78%	NA	
Keener et al. ¹³	146° (NA)	ASES: 74 (NA) SST: 8.9 (3.5) VAS: 2.7 (NA)	52%	0% Surgical complication
Piasecki et al. ⁷⁵	136° (15)	ASES: 68 (24) SST: 7.5 (4) VAS: 2.7 (2.4)	NA	11.1%

ASES, American Shoulder and Elbow Surgeons; CS, constant score; NA, not available; SST, simple shoulder test; UCLA, University of California Los Angeles; VAS, visual analogue scale.

following revision RTC surgery has been reported as high as 53% at the 2-year follow-up.⁴⁴ When it does occur, retearing has been shown to have a poor prognosis with regard to patient function and pain relief.⁷¹ These rates are the reason that patient selection and proper preoperative counseling are imperative for revision cuff repair. The failure of a revision cuff repair should be approached with extreme caution as an increased number of surgeries have been shown to directly correlate with an increased rate of complications and failures.²

FUTURE CONSIDERATIONS

Further investigation into biologic augmentation methods will have the greatest effect on the future of revision cuff surgery. Gene therapy and stem cell therapy could have a tremendous effect on optimizing the biology of the degenerating shoulder and help to prevent primary cuff failure and to improve the outcome of revisions.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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